

VISHNU BHUVANACHANDRAN

Innovation and entrepreneurship development officer

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Summary

An R&D professional with over 10 years of experience in systems design and prototyping, expert in SolidWorks and 3D printing. Key achievements include boosting research capacity by 80% and earning a President's Gold Medal in Data Analytics. Seeking a Senior Mechanical Design Engineer position at Stryker, where skills in systems design and prototyping will be brought to support the mission of designing and delivering next-generation interventional solutions.

Experience

Marian Engineering College Kerala, India

Research lead & Assistant Nodal Officer (IEDC) 09/2019 - 08/2024

- Oversaw and reviewed the project operations, drafted intellectual property documentations and coordinated the startup incubations.
- Managed a rapid prototyping lab (3D printing/CNC/laser), accelerating modelling, iteration cycles and retiring technical risks early in the development process. Consulted projects.
- Launched Energy Research Lab and Makers Lab, boosting research and outsourcing capacity by 80% through detailed proposals and supplier negotiation.
- Encouraged and coached students through hands-on prototyping, increasing their confidence in modern manufacturing techniques and lab use.
- Delivered innovation training to students, transforming ideas into startups and streamlining documentation processes.

Indira Gandhi Centre for Atomic Research Tamilnadu, India

Senior Research Fellow 05/2014 - 12/2019

- Accomplished 2 research projects with Govt. of India. Designed, developed and validated components under safety standards, executed DoE-based experiments. Co-led a green facility setup for low-emission consumables.
- Built and tested prototypes, performed failure analysis and communicated findings to cross-functional teams, implementing corrective actions through structured change processes and supporting audits and compliance.
- Authored documentation, risk assessments, patents, and publications. Collaborated with stakeholders. Prepared successful proposals using strong interpersonal skills.

Education

Dundalk Institute of Technology Ireland

MSc, Data Analytics - President's Gold Medal Winner 2024 - 2025

NIT Tiruchirappalli India

PhD, Mechanical Engineering (Occupational Health and Safety) 2014 - 2020

University of Kerala India

M.Tech, Thermal Science 2011 - 2013

University of Kerala India

B.Tech, Mechanical Engineering 2005 - 2009

Skills

Project Management • Continuous learning • Teamwork • 3D Modeling • Analyzing data • Auto-CAD • Solidworks • Creo • FEA • ANSYS • Problem-solving • Communication • Data visualization • Presentation skill • Microsoft Office • Spreadsheets • Word Processing • Minitab • Python • Neo4j • SQL • Machine Learning • 3D Printing • CNC

Projects

Selected Projects

- Strategy for reduction of carcinogenic welding emissions resulting in green welding consumable development with optimized nano based flux compositions aiming better occupational health.
- EV Battery Thermal Management: CAD, stack-up, material selection, prototyping, and verification notes; applied DFM/DFA to simplify assembly and minimise risk.
- Central Vacuum System (Industrial): Supplier collaboration for components and BoM; manufacturable drawings; revision-controlled packs; timely responses to engineering info/drawing requests.
- Industrial development Project for developing and uplifting Handloom cluster of the Kerala State- Socio-economic Development.
- Built a machine learning pipeline for early cattle disease prediction from a 2.3M-row dataset. Applied feature engineering, class-imbalance techniques, and model optimization for data-driven health decisions.

Awards and Fellowships

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|  First Class Honors in M.Sc. Data Analytics from Dundalk Institute of Technology - President's Award Winner |  Senior Research Fellowship - Indira Gandhi Centre for Atomic Research |  Best Paper Award-Dual Stage Thermoelectric Cooling Method for EV batteries |
|  Spoke on IIW assemblies addressing the impacts of Welding Emissions and importance of Green Welding processes | | |

Publication and Patents

4 Granted, 1 Published (consumables/thermal systems/safety devices). Titles include welding flux compositions, thermoelectric EV battery cooling concept, fume-reduction helmet, and thermal-spray attachments. Full list available on request.

15+ publications across peer-reviewed journals and conferences (design, process safety, and manufacturing). Full list available on request.

Standards, Certifications and Upskilling

ISO 13485 — Udemy - Quality Management System for Medical Devices

PLM Fundamentals (Udemy, Beginner Certificate) — items/BOM, revisions, release workflows, ECR/ECO concepts

ISO 15011-1:2009, 10882-1, AWS F1.2 — Welding Emission Characterization and associated set up development

Basic understanding of IEC 61439 & UL 857 — Electrical product design standards for low-voltage switchgear and busbar trunking systems (self-study, 2026)

Advanced MS Office -Word, Spread sheets, PowerPoint — Skill Nation

Advanced AI hacks with MS Office — Skill Nation

Advanced Microsoft Power BI — Skill Nation